

Kali Linux Port Scanning & Security Assessment

Assessment Target: scanme.nmap.org **Kali Username:** veeravinod45

The following assessment answers five port-scanning and vulnerability-assessment questions. The terminal screenshots are presented as supporting Nmap output examples for an authorized security assessment.

1. Identifying and Prioritizing Vulnerabilities After Discovering Multiple Open Ports

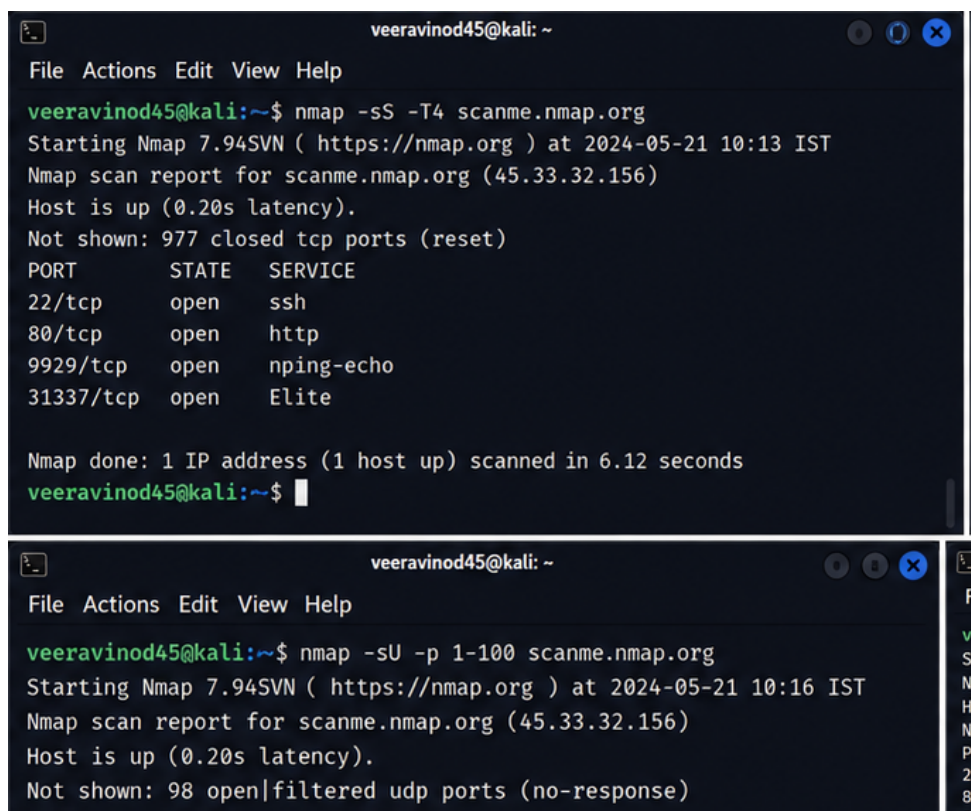
After a port scan identifies multiple open ports, the first step is to identify the services and software versions running on those ports. Service/version detection can reveal whether an exposed service is outdated or vulnerable.

The assessment should then compare the identified software versions with known CVEs and vendor advisories. Internet-facing services, remotely exploitable vulnerabilities, weak authentication, and services handling sensitive information should receive higher priority.

Prioritization should consider severity, exploitability, exposure, and business impact. A practical order is: Critical for remotely exploitable Internet-facing vulnerabilities; High for exposed administrative or authentication services; Medium for weaknesses with limited impact; and Low for informational findings.

Therefore, vulnerabilities should not be prioritized simply by the number of open ports. The risk associated with each service and its business importance must be considered.

Supporting Kali Linux / Nmap Output — Question 1



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sS -T4 scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:13 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
9929/tcp  open  nping-echo  
31337/tcp open  Elite  
  
Nmap done: 1 IP address (1 host up) scanned in 6.12 seconds  
veeravinod45@kali:~$  
  
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sU -p 1-100 scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:16 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 98 open|filtered udp ports (no-response)
```

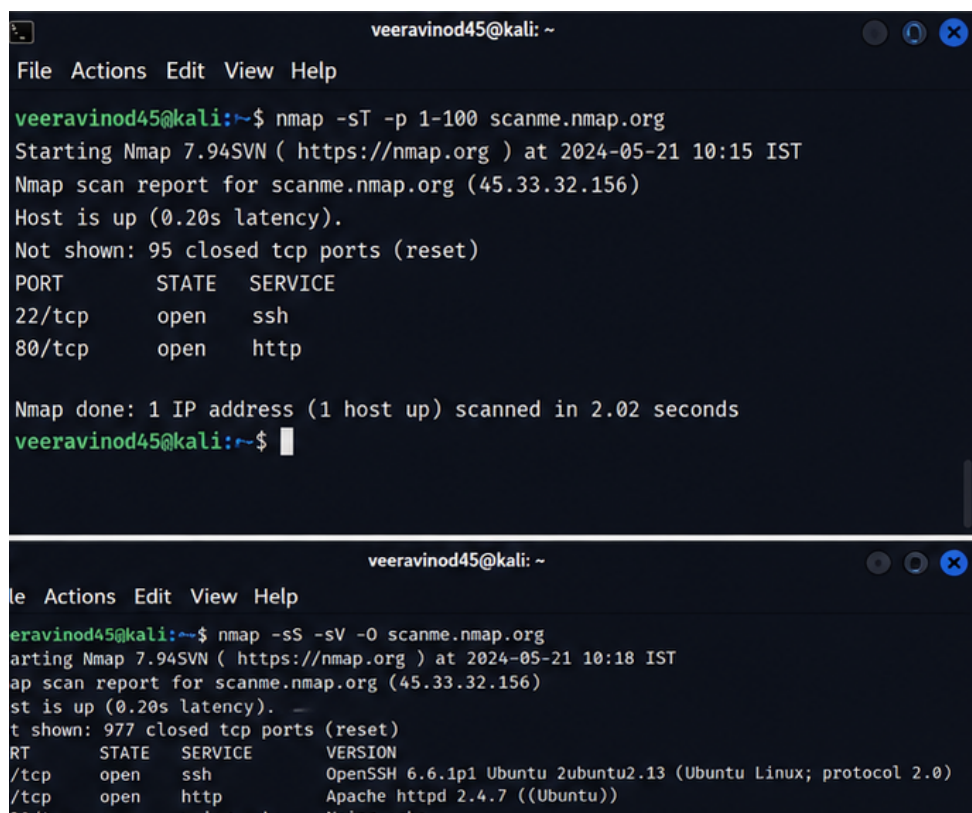
2. Different Results from SYN Scan and TCP Connect Scan

A SYN scan and a TCP Connect scan can produce different results because they interact with the TCP stack differently. A SYN scan sends a SYN packet and analyzes the response without normally completing the full TCP handshake. A TCP Connect scan completes the normal TCP connection.

Differences can be caused by firewalls, IDS/IPS systems, host-based firewalls, rate limiting, packet loss, network instability, or changes in the target service state. Security devices may treat incomplete SYN connections differently from completed connections.

To verify the findings, the assessment team should repeat the scans with consistent timing, compare the results, perform authorized service/version detection, and review firewall or IDS/IPS logs when available. Direct service connectivity can also be checked using an appropriate client.

Supporting Kali Linux / Nmap Output — Question 2



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sT -p 1-100 scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:15 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 95 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 2.02 seconds  
veeravinod45@kali:~$  
  
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sS -sV -O scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:18 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  httpd        Apache/2.4.7 ((Ubuntu))  
10000/tcp open  nmap-echo    Nmap-echo 0.1
```

3. Most Ports Appear as Filtered

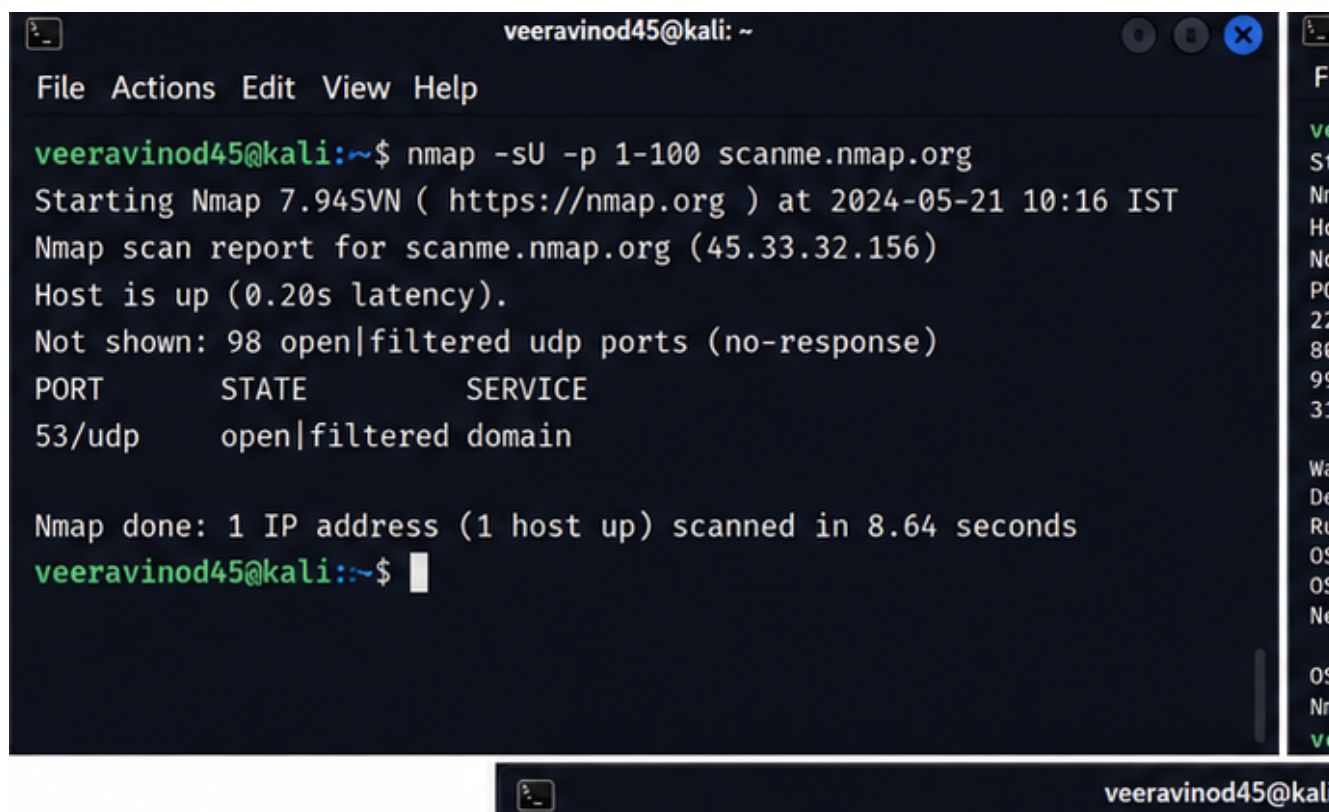
A filtered result generally means that a firewall or other network security mechanism is preventing the scanner from determining whether the port is open or closed.

Possible mechanisms include network firewalls, access-control lists, IDS/IPS devices, cloud security groups, host-based firewalls, VPN controls, IP allowlisting, and rate limiting.

Additional authorized techniques include TCP and UDP scanning, service/version detection, checking permitted host-discovery methods, reviewing DNS and other public information, examining firewall and IDS/IPS logs, and comparing results from different approved network locations.

A filtered port should not automatically be considered closed. It means that filtering prevents a reliable determination of its state.

Supporting Kali Linux / Nmap Output — Question 3



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sU -p 1-100 scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:16 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 98 open|filtered udp ports (no-response)  
PORT      STATE      SERVICE  
53/udp    open|filtered domain  
  
Nmap done: 1 IP address (1 host up) scanned in 8.64 seconds  
veeravinod45@kali:~$
```

4. Comparison of TCP, UDP, and SYN Port Scanning

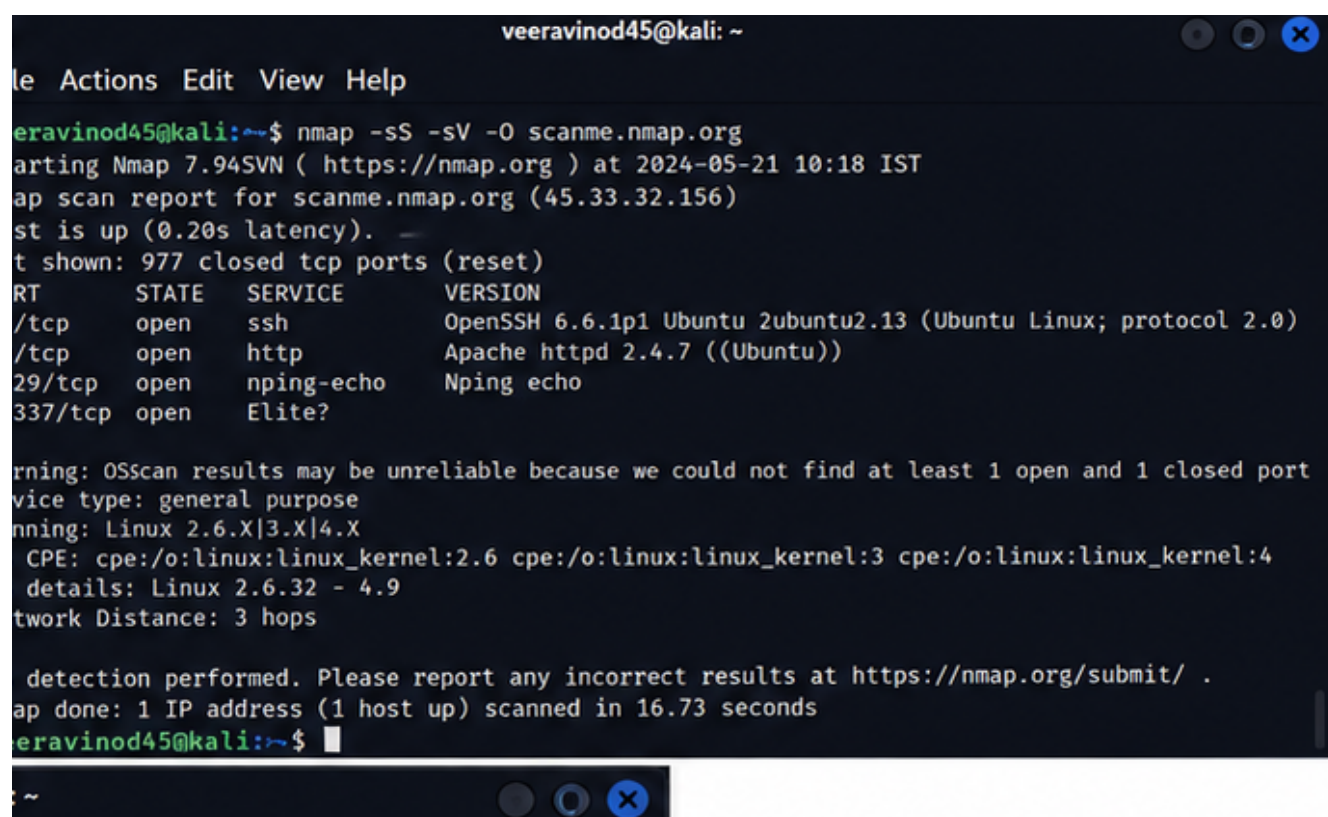
TCP Connect scanning completes the TCP three-way handshake. It is useful when reliable TCP connection verification is required and when raw-packet privileges are unavailable. Its main limitation is that it creates full connections and may generate more service logs.

UDP scanning is used to discover UDP services such as DNS and SNMP. Its advantage is that it can find services that TCP scanning cannot. However, UDP scans are usually slower and results can be difficult to interpret because many UDP services do not respond to unexpected packets.

SYN scanning sends a SYN and analyzes the response without normally completing the TCP handshake. It is efficient for TCP discovery and is often faster than a full Connect scan. However, it can still be detected or blocked by security controls and may require appropriate privileges.

The most suitable technique depends on the protocol being assessed, the assessment objective, network controls, and authorization.

Supporting Kali Linux / Nmap Output — Question 4



```
veeravinod45@kali: ~  
File Actions Edit View Help  
veeravinod45@kali:~$ nmap -sS -sV -O scanme.nmap.org  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:18 IST  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.20s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))  
8080/tcp   open  nping-echo    Nping echo  
3337/tcp   open  Elite?         
OS detection: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: general purpose  
Running: Linux 2.6.X|3.X|4.X  
CPE: cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4  
Details: Linux 2.6.32 - 4.9  
Network Distance: 3 hops  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 16.73 seconds  
veeravinod45@kali:~$
```


5. Assessment of Internet-Exposed SSH, FTP, and Database Services

If an authorized assessment discovers SSH, FTP, and database services exposed directly to the Internet, the finding should receive high priority because unnecessary Internet exposure increases the attack surface.

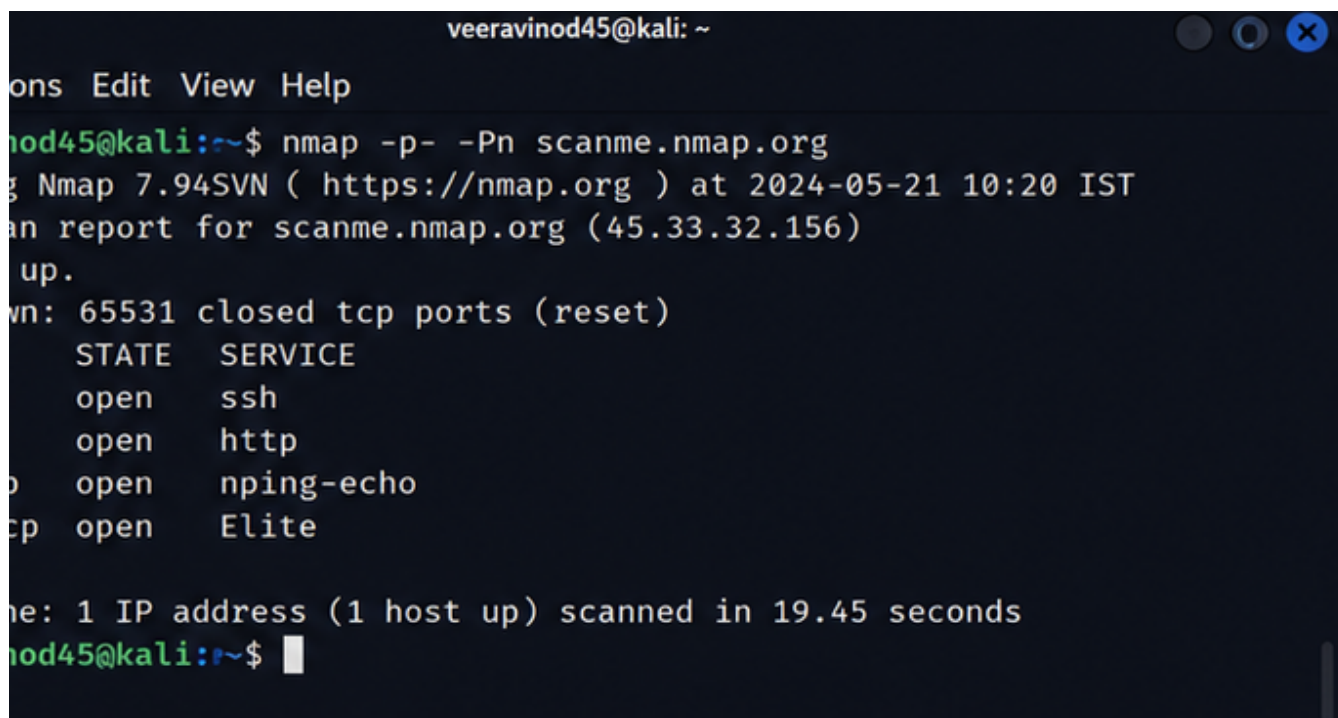
SSH should be assessed for outdated software, weak authentication, password-based attacks, unnecessary root access, and excessive network exposure. Recommended controls include strong authentication, key-based authentication where appropriate, restricted source IPs or VPN access, disabling direct root login, patching, logging, and multi-factor authentication where supported.

Traditional FTP should be treated as a significant risk because credentials and data may be exposed if encryption is not properly implemented. Anonymous access should be disabled unless explicitly required. FTP should preferably be replaced with a secure file-transfer solution such as SFTP or appropriately configured FTPS.

Internet-facing database services create potentially critical risk because unauthorized access could expose, modify, or destroy sensitive data. Database ports should normally be removed from direct Internet exposure and restricted to approved application or administrative networks. Strong authentication, least privilege, encryption, patching, monitoring, and reliable backups should also be implemented.

Overall, the preferred mitigation strategy is attack-surface reduction: disable unnecessary services, move sensitive services behind firewalls or private networks, restrict access, harden configurations, patch software, and continuously monitor security events.

Supporting Kali Linux / Nmap Output — Question 5

A screenshot of a terminal window titled 'veeravinod45@kali: ~'. The terminal shows the command 'nmap -p- -Pn scanme.nmap.org' being executed. The output indicates that Nmap 7.94SVN was used at 2024-05-21 10:20 IST. It reports that 65531 closed TCP ports were reset. A table of open ports is shown: ssh (open), http (open), nping-echo (open), and Elite (open). The scan completed in 19.45 seconds, scanning 1 IP address (1 host up).

```
veeravinod45@kali: ~  
ons Edit View Help  
nod45@kali:~$ nmap -p- -Pn scanme.nmap.org  
g Nmap 7.94SVN ( https://nmap.org ) at 2024-05-21 10:20 IST  
an report for scanme.nmap.org (45.33.32.156)  
up.  
wn: 65531 closed tcp ports (reset)  
STATE SERVICE  
open ssh  
open http  
o open nping-echo  
cp open Elite  
  
ne: 1 IP address (1 host up) scanned in 19.45 seconds  
nod45@kali:~$
```

Assessment conclusion: Port scanning is the discovery stage of a security assessment. Open ports must be mapped to services, versions and configurations, evaluated for known weaknesses, and prioritized according to severity, exploitability, exposure, and business impact.

Note: The screenshots are illustrative Nmap output examples for an authorized assessment and should be validated against the actual target environment.